

PHY-275 Fall 2026 Assignment #5 due 10/7/26 9:00pm

You will receive 1 credit upon complete all the tasks.

Task #1: Practice Taylor expansion, work out problem 2.18 from your textbook, write down all the necessary steps.

Task #2: the Coriolis force

On Wednesday we are going to derive Coriolis force in the Cartesian coordinate system, however our J. R. Taylor expressed it in a slightly different form at equation (9.49):

$$F_{cor} = 2m\dot{\vec{r}} \times \vec{\omega}$$

Now apply Lagrangian mech to derive the Coriolis force in **polar coordinates**, write down all the necessary steps. Give a quick read of chapter 9.6 if necessary.

To proof your efforts, write down each step and upload them to the Moodle site as a single file.